



DIGESTION AND METABOLISM

IMCQ 6 ANAT and HCB

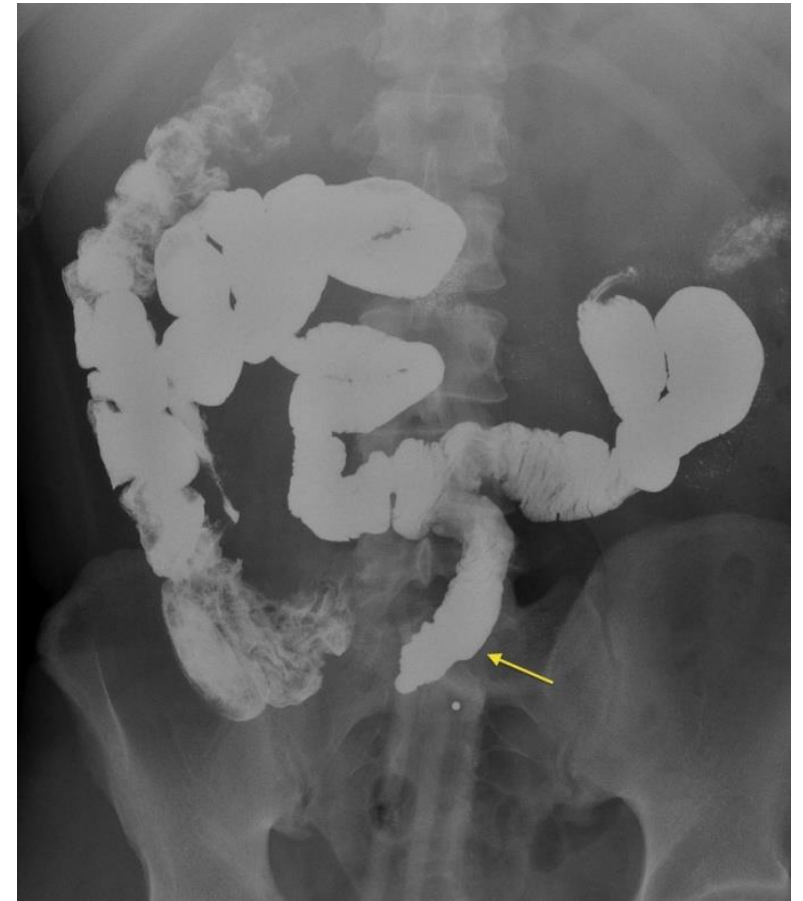
1. A 60-year-old woman comes to the emergency department because of a 2-day history of fever and severe abdominal pain. Physical examination shows muscle guarding and rebound tenderness on deep palpation. An x-ray and barium swallow of her abdomen is shown. Which of the following groups of lymph nodes most likely drains the part of the GI indicated by the arrow?



- A. Internal iliac
- B. Inferior mesenteric
- C. Lumbar
- D. Celiac
- E. Inferior mesenteric
- F. Superior mesenteric

2. A 25-year-old man comes to the emergency department because of a 3-hour history of severe abdominal pain. Physical examination shows guarding whenever the abdomen was touched. A barium swallow is done which is shown in the image. There is inflammation of the structure at the tip of the arrow. Which of the following is the most likely diagnosis?

- A. Congenital megacolon
- B. Meckel's diverticulum
- C. Omphalocele
- D. Hypertrophic pyloric stenosis
- E. Omphaloenteric fistula
- F. Anorectal agenesis with fistula



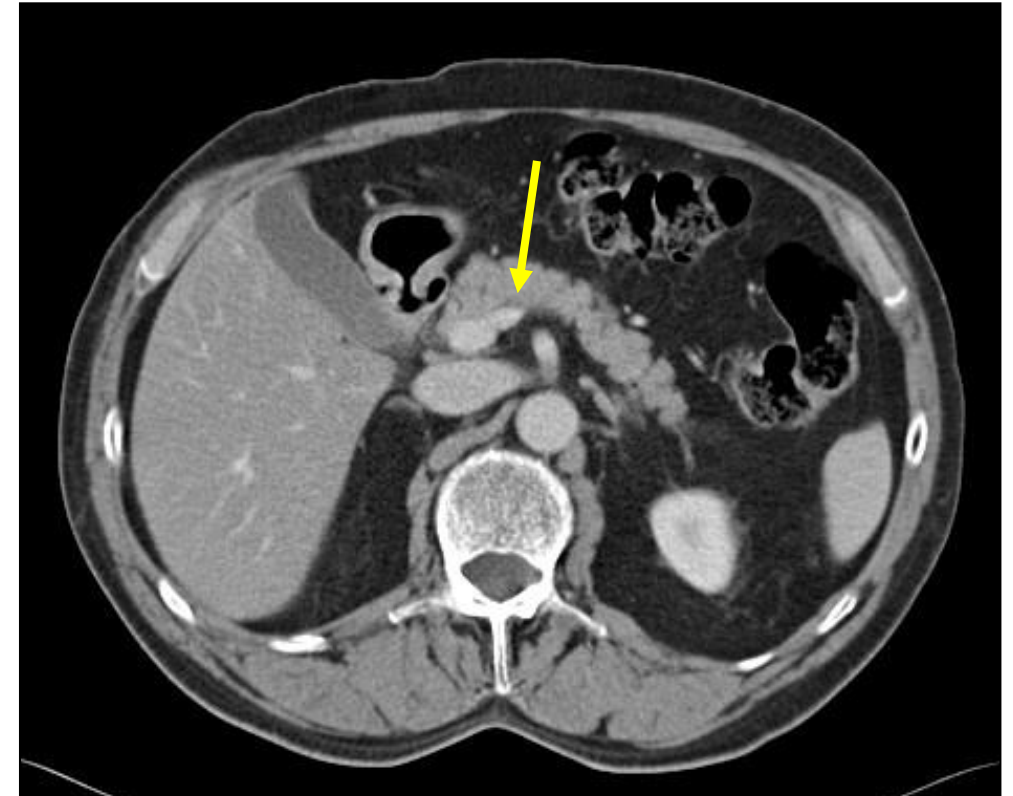
3. A 70-year-old man is brought to the emergency department by his daughter because of a 1-day history of abdominal pain, nausea and vomiting. He has a history of constipation. Physical examination shows a distended abdomen. A radiograph of his abdomen is shown in the attached image. Which of the following arteries most likely directly supplies the part of the GI that is affected?

- A. Middle colic
- B. Superior rectal
- C. Sigmoidal
- D. Right colic
- E. Middle rectal
- F. Jejunal

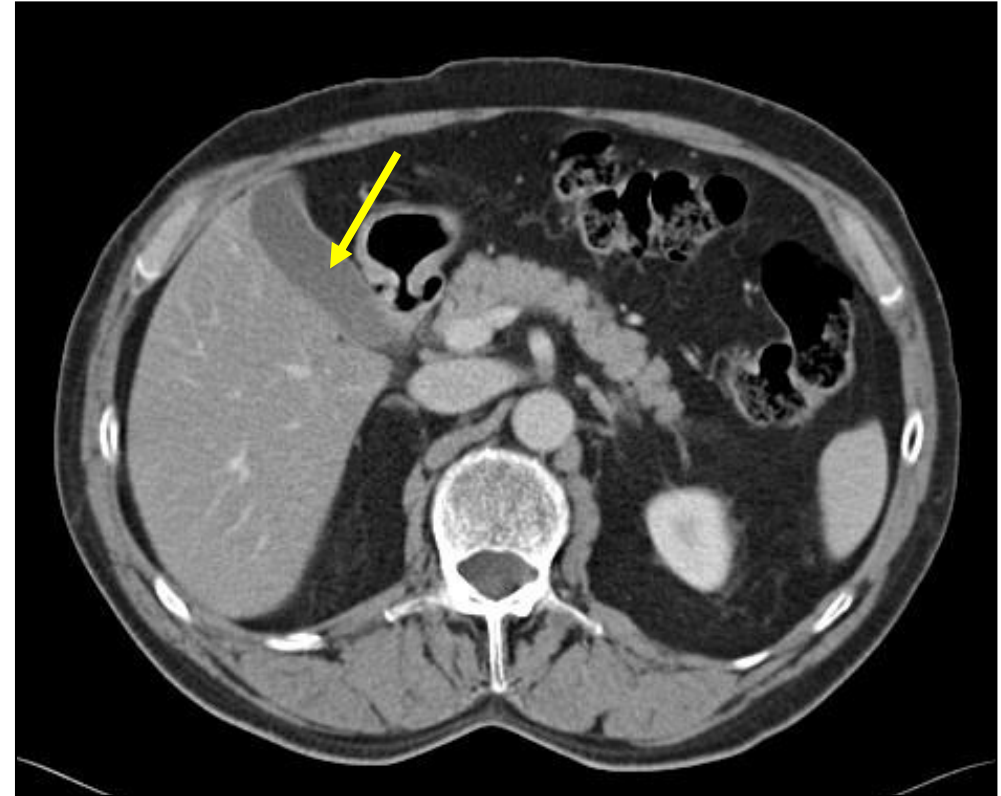


4. A 36-year-old man comes to the emergency department because of a 6-hour history severe abdominal pain that radiates to his back. His temperature is 39°C (103°F). Physical examination shows rebound tenderness. Laboratory studies show increased levels of serum amylase and lipase. He is diagnosed with inflammation of the organ at the tip of the arrow in the attached image. Which of the following vessels most likely runs on the superior aspect of the affected organ?

- A. Gastrooduodenal
- B. Common hepatic
- C. Left gastro-omental
- D. Splenic
- E. Left gastric
- F. Right gastro-omental



5. A 15-year-old girl is brought to the emergency department because of a 4-hour history of abdominal pain and vomiting. Physical examination shows tenderness in the epigastric and right hypochondrium. Imaging studies show thickening and inflammation of the walls of the organ at the tip of the arrow. Which of the following nervous pathways most likely carries the pain fibers from the affected organ back to the spinal cord?



- A. Preganglionic parasympathetics in vagus
- B. Postganglionic parasympathetics in pelvic splanchnic
- C. Preganglionic sympathetics in greater thoracic splanchnic
- D. Postganglionic sympathetic in lesser thoracic splanchnic
- E. Postganglionic sympathetics in the inferior mesenteric periarterial plexus
- F. Postganglionic sympathetic in greater thoracic splanchnic

6. A 40-year-old woman is brought to the emergency department unresponsive after a crash. Her blood pressure is 85/60mmHg, respirations are 20/min and pulse is 110/min. Fluid resuscitation is started with two large bore needle catheters placed in the antecubital fossa of her upper limbs. Physical examination shows extensive abdominal bruising in the right upper quadrant. CT scan of the abdomen shows a liver laceration and abdominal surgery is scheduled. Which of the following is most likely the embryological origin of the peritoneal ligament that is ligated during surgery to control the bleeding?

- A. Ventral mesentery
- B. Midgut
- C. Hindgut
- D. Dorsal mesentery
- E. Foregut

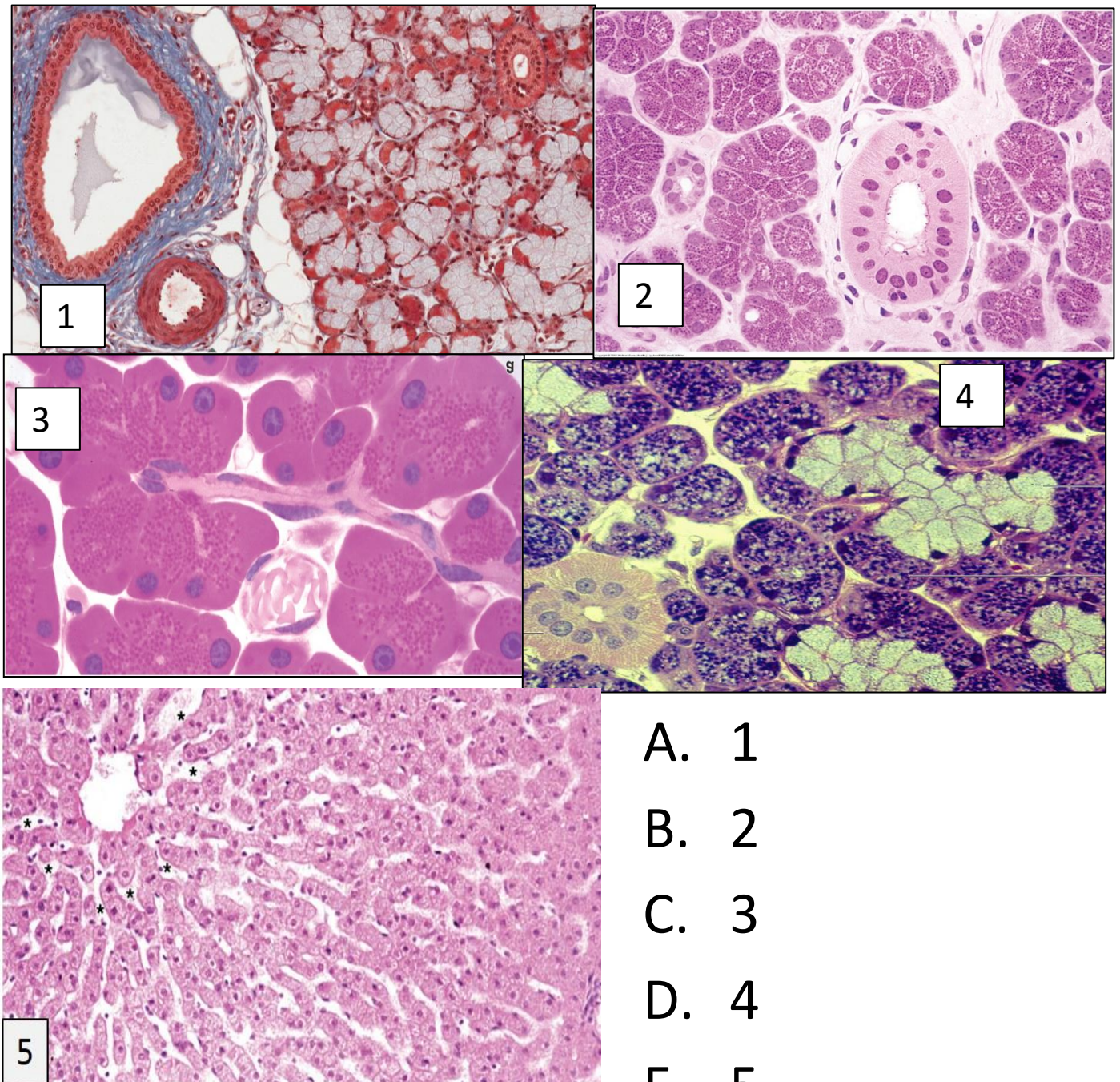
7. A 60-year-old man is brought to the emergency department by his wife because of a 12-hour history of passing bloody stools. He has a 10-year history of alcoholism and has been previously treated for hemorrhoids and jaundice. Physical examination shows dilated veins radiating from his umbilicus. Rectal exam shows the stool contains bright red blood. Which of the following veins most likely forms the caval part of the anastomosis that is responsible for the patient's presenting complaint?

- A. Left gastric
- B. Superior rectal
- C. Inferior rectal
- D. Paraumbilical
- E. Superficial epigastric
- F. Sigmoidal

8. A 4-day-old girl is admitted to the hospital because of a 2-day history of high fever and refusal to feed. Physical examination shows abdominal distention and meconium (first stool) coming from the vagina. Imaging studies show a connection between the rectum and the vagina. Which of the following is the most likely diagnosis?

- A. Anorectal agenesis
- B. Anal stenosis
- C. Membranous atresia
- D. Omphaloenteric fistula
- E. Duodenal stenosis
- F. Congenital megacolon

9. A 35-year-old man comes to the emergency department because of a 2-hour history of epigastric abdominal pain that radiates to the back. His blood pressure is 100/60 mm Hg, pulse is 100/min, respirations are 22/min and temperature is 38.3°C (101°F). Physical examination shows a diaphoretic patient with abdominal wall guarding and rebound tenderness. Laboratory studies show elevated serum lipase and amylase. Which of the following images is the histological slide of the organ that is most likely affected?



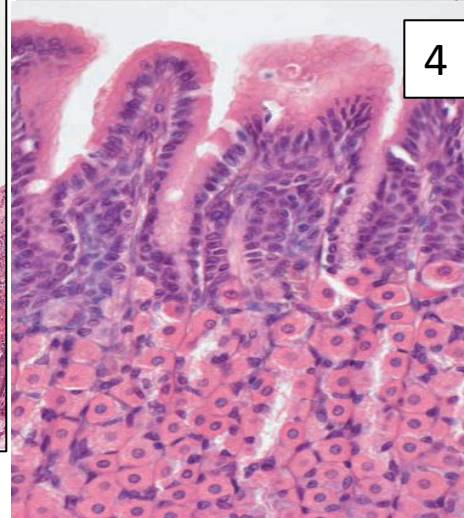
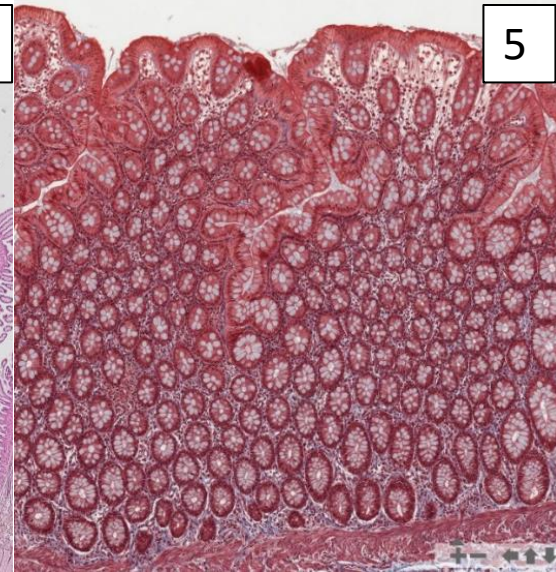
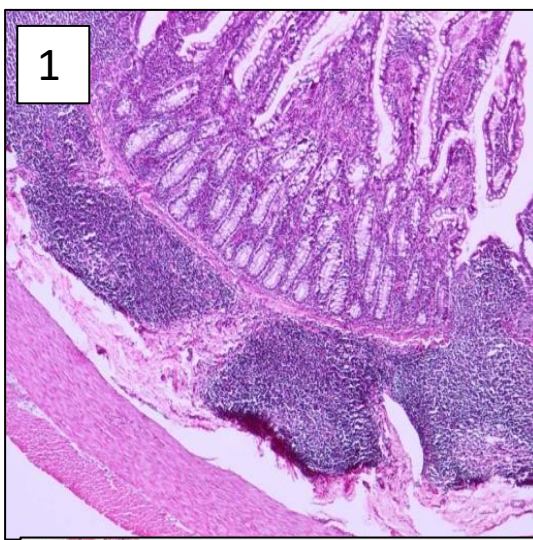
- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

10. A 38-year-old woman comes to the emergency department because of a 3-day history of nausea, vomiting and intermittent “colicky” abdominal. She says her stool is “clay-colored” and her urine is dark. Physical examination shows an ill looking patient with yellowing of the skin. There is also right upper quadrant tenderness on deep palpation. Laboratory tests show elevated total bilirubin, AST, ALT and ALP enzymes. Lipase and amylase enzymes are within normal range. Ultrasound shows large gallstones in the biliary ductal system. Which of the following cells lines the structure that is most likely blocked by the gallstones that explain this patient’s clinical presentation?

- A. Hepatocytes
- B. Serous acinar cells
- C. Kupfer cells
- D. Columnar absorptive cells
- E. Ito cells
- F. Cholangiocytes
- G. Sinusoidal endothelial cells

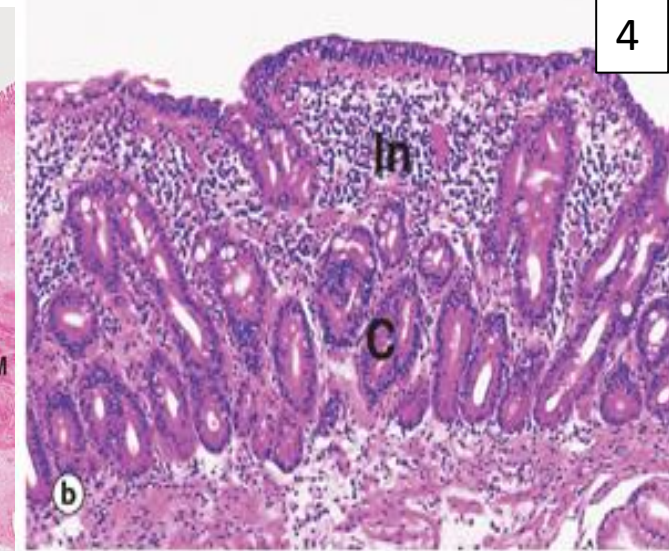
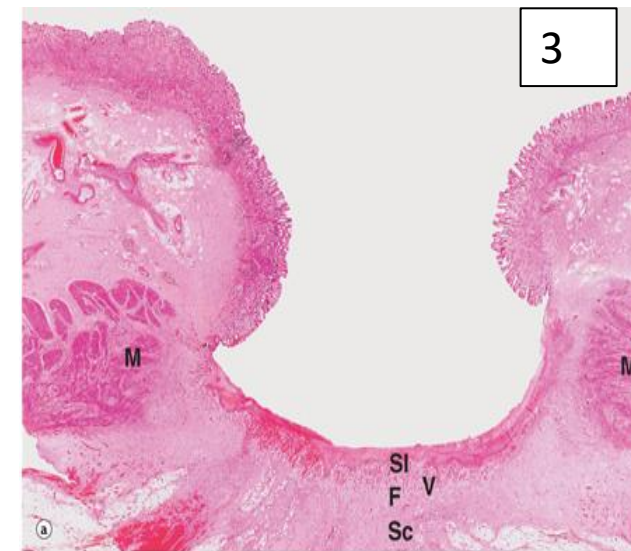
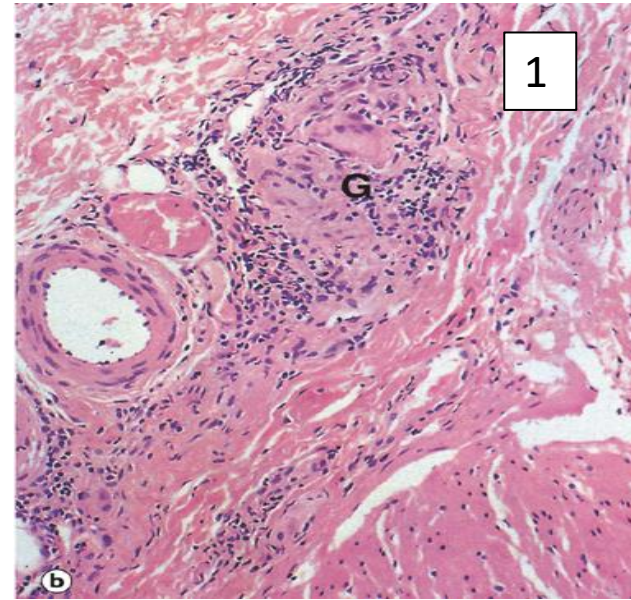
11. A 28-year-old woman comes to the physician because of a 1-year history of recurrent diarrhea, abdominal pain, fever and malnutrition. Endoscopy shows inflammation of the part of the bowel located 2 feet (~60cm) distal to the stomach. Which of the following is most likely the area affected?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5



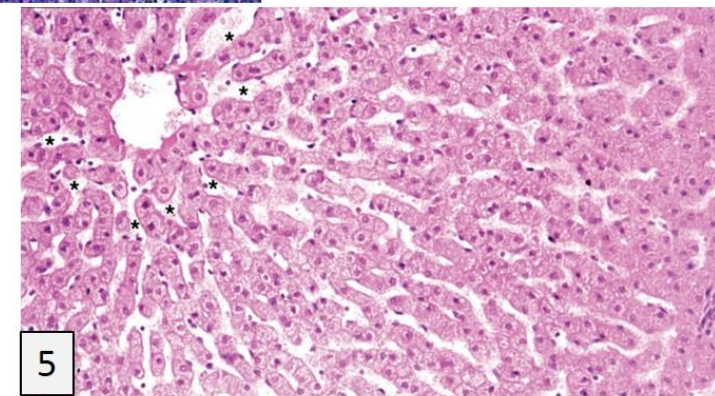
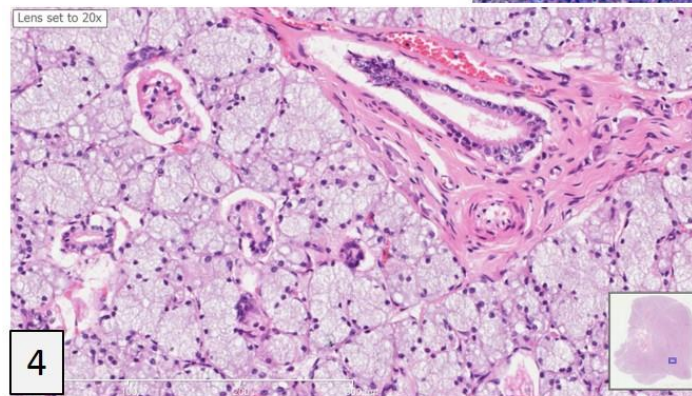
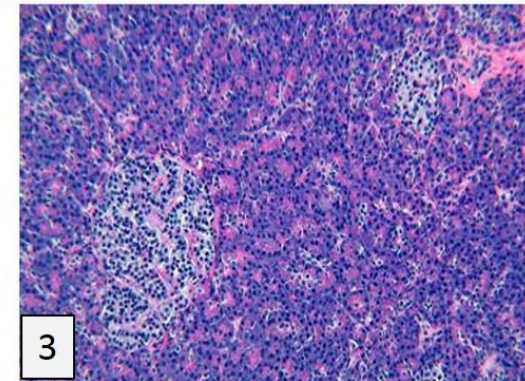
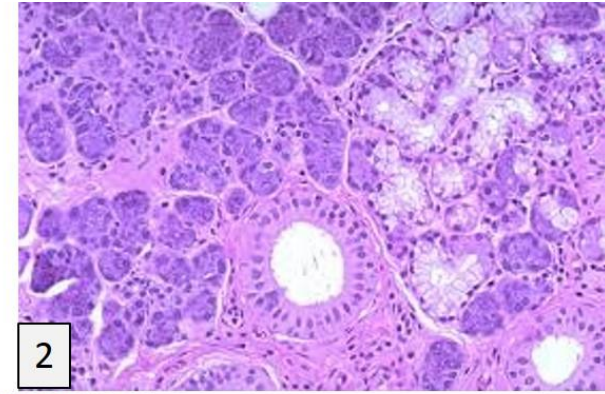
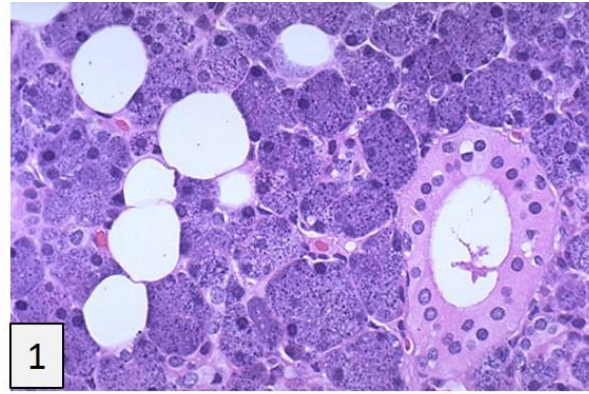
12. A 35-year old woman comes to the physician because of a 1-day history of flatulence, abdominal pain and diarrhea. The symptoms began after eating a heavy meal the night before. She has a 15-year history of bowel frequency with occasional bouts of constipation which has gotten worse over the past 2 years. She has also lost a significant amount of weight over the same time period. Physical examination shows a thin, pale and ill-looking patient with sunken eyes and scanty hair. Laboratory studies shows deamidated gliadin peptide IgA antibodies. Which of the following will most likely be seen in a biopsy of the affected part of the GI system?

- A. 1
- B. 2
- C. 3
- D. 4

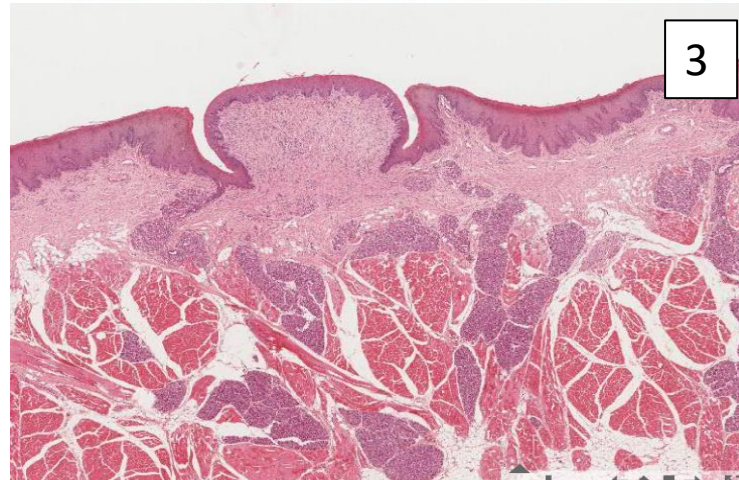
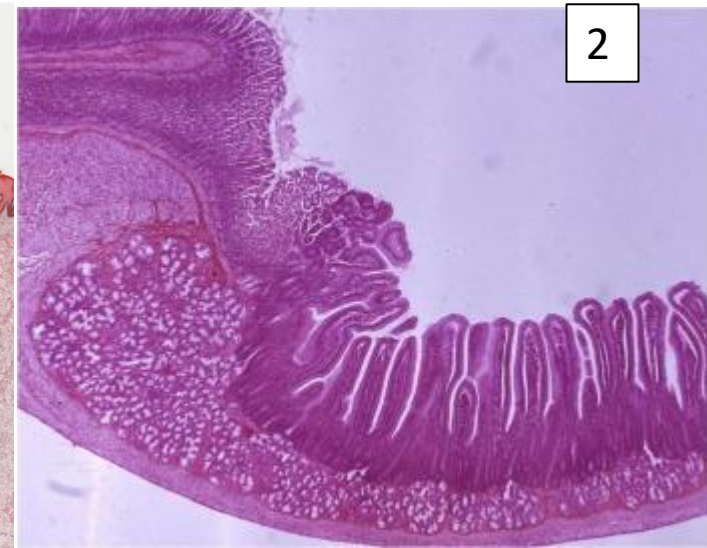
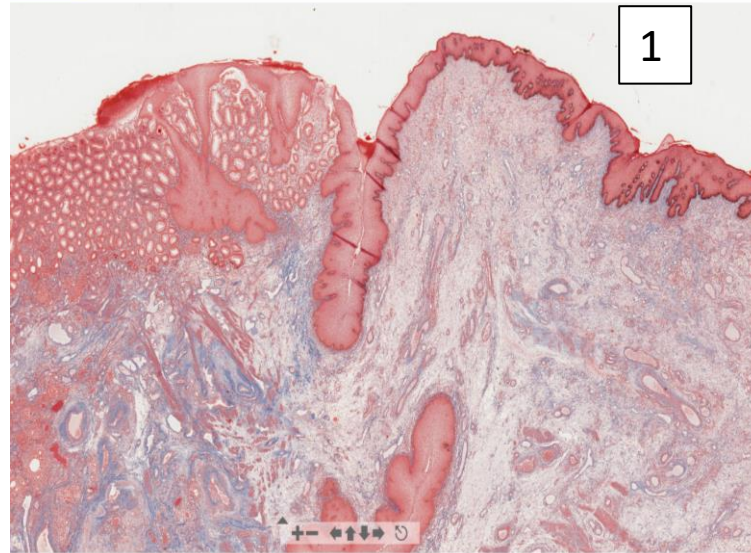


13. A 64-year-old man comes to the physician because of a 3-month history of a swelling in the floor of the mouth. Physical examination shows a firm, 3 cm mass in sublingual salivary gland. Specimens are shown in the attached images. Which of the following images best corresponds to the affected gland?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5



14. A 57-year-old woman comes to the physician because of a 6-month history of irregular bowel habits and blood in the stool. Physical examination shows abdominal tenderness and macroscopic rectal bleeding. Laboratory studies of the blood show no abnormalities. Colonoscopy shows a 1.5 cm mass in the distal part of the GI tract at anal canal. Which of the following images best corresponds to this region?

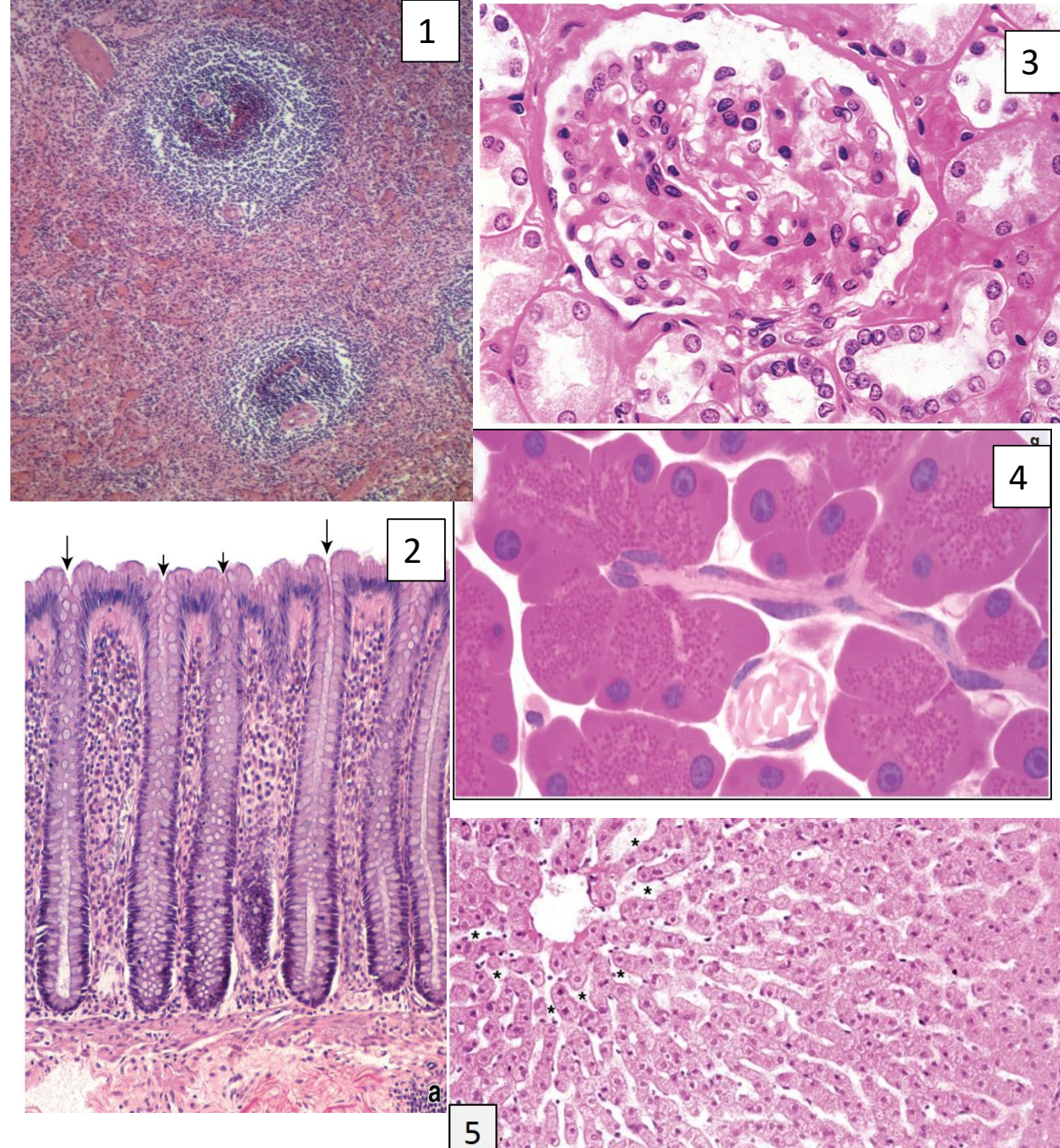


- A. 1
- B. 2
- C. 3
- D. 4

15. A 37-year-old man comes to the emergency department because of a 1-hour history of a deep puncture wound to the abdomen while trying to climb a barbed wire fence. Physical examination shows the wire had punctured the anterior abdominal wall and has entered one of the abdominal organs, leading to the release of fluid and air into the abdominal cavity. A radiograph is taken which shows air in the subphrenic recess. Which of the following is the organ that is most likely punctured?

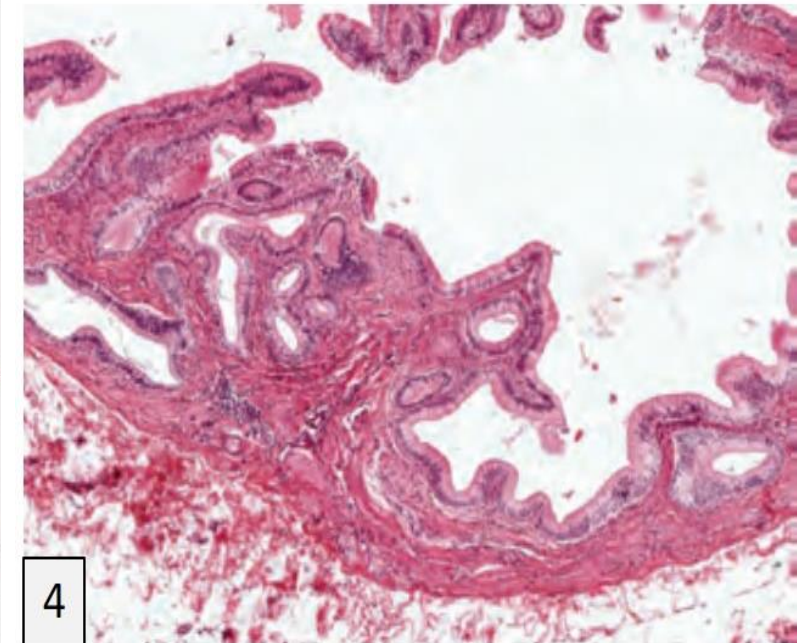
- A. 1
- B. 2
- C. 3
- D. 4
- E. 5

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16. A 53-year-old man comes to the emergency department because of a 4-hour history of severe abdominal pain and nausea. The pain started around the umbilical region and now intensifies and shifted to the right iliac fossa. Physical examination shows positive psoas test and rebound tenderness at McBurney's point. Which of the following slides best describes the inflamed organ?

- A. 1
- B. 2
- C. 3
- D. 4



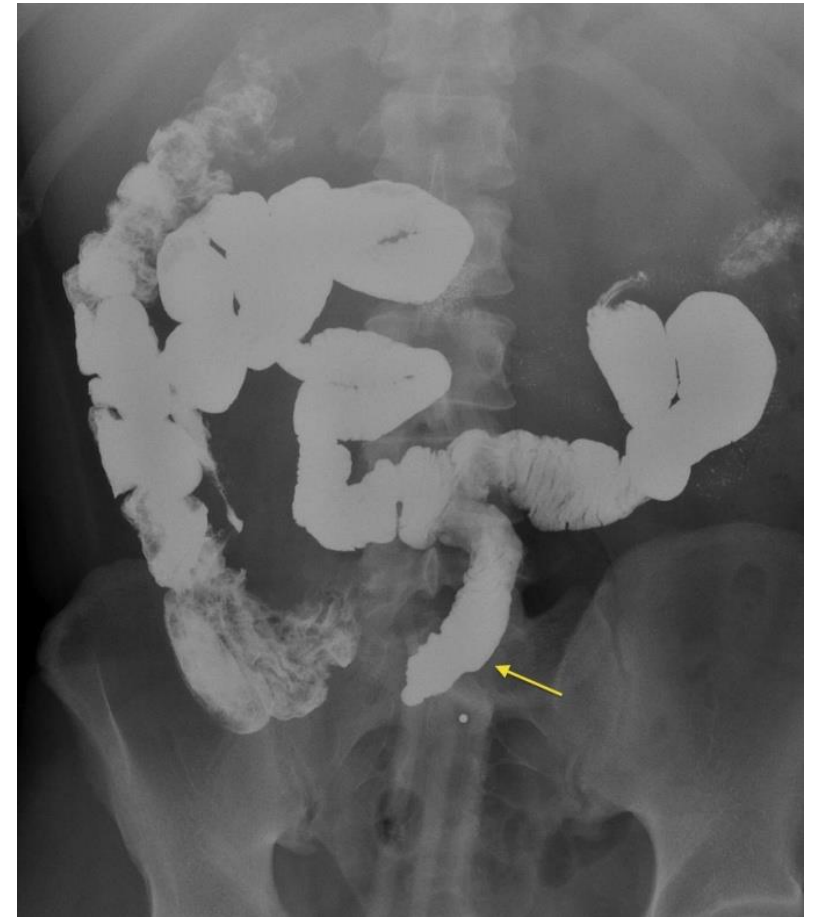
17. A 60-year-old woman comes to the emergency department because of a 2-day history of fever and severe abdominal pain. Physical examination shows muscle guarding and rebound tenderness on deep palpation. An x-ray and barium swallow of her abdomen is shown. Which of the following nerves most likely brings sympathetic innervation to the part of the GI indicated by the arrow?



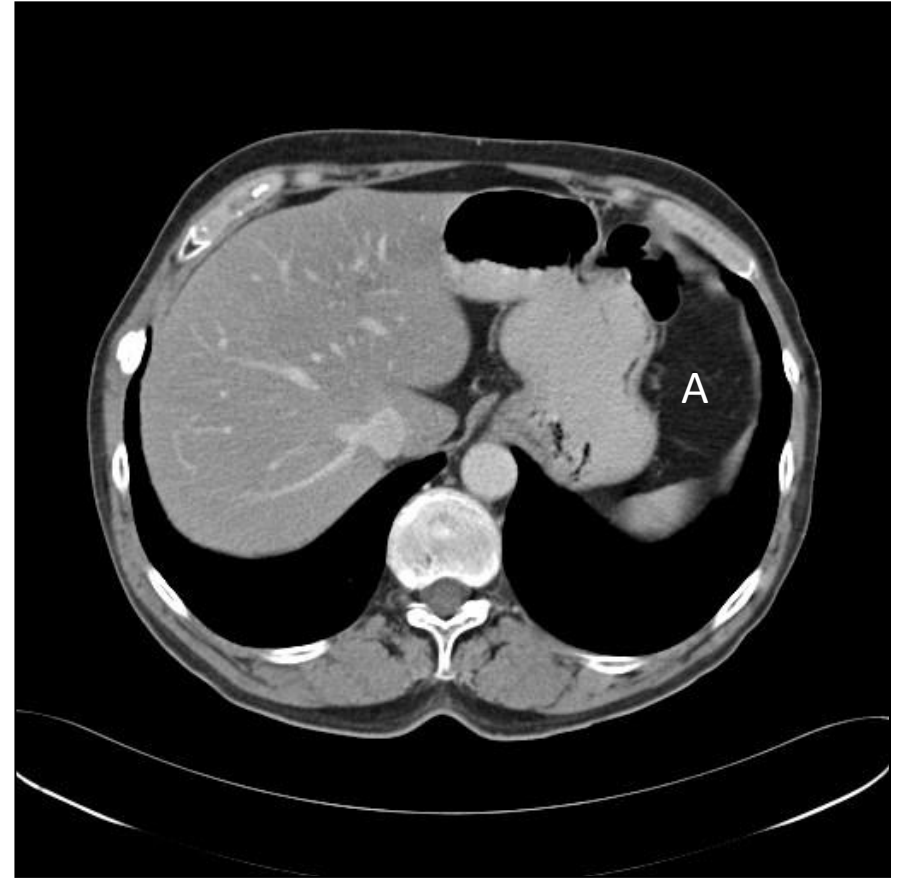
- A. Lumbar splanchnics
- B. Pelvic splanchnics
- C. Lesser thoracic splanchnics
- D. Greater thoracic splanchnics
- E. Vagus

18. A 25-year-old man comes to the emergency department because of a 3-hour history of severe abdominal pain. Physical examination shows guarding whenever the abdomen is touched. A barium swallow is done which is shown in the image. There is inflammation of the structure at the tip of the arrow. Which of the following types of ectopic tissue will most likely be found in this structure?

- A. Pancreatic and hepatic
- B. Gastric and hepatic
- C. Salivary and gastric
- D. Gastric and pancreatic
- E. Salivary and pancreatic



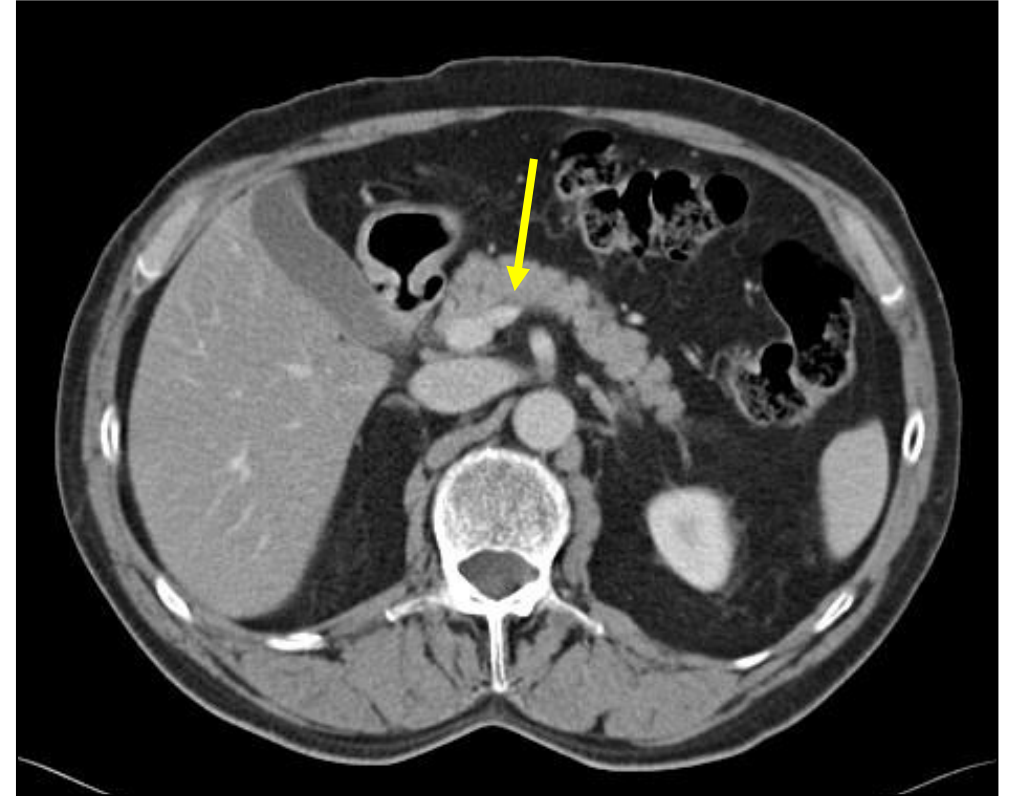
19. A 75-year-old woman comes to the emergency department because a 2-day history of nausea, vomiting, intense abdominal pain and inability to pass stool. She has had unintentional weight loss of 15lbs (6.8 kg) over the last three months. Physical examination shows a distended abdomen. Imaging studies show cancer of the colon extending from the left colic flexure to the middle of the descending colon. There is metastasis to the structure labeled A in the attached image. Which of the following is most likely the embryological origin of the labeled structure?



- A. Ventral mesentery
- B. Midgut
- C. Hindgut
- D. Dorsal mesentery
- E. Foregut

20. A 36-year-old man comes to the emergency department because of a 6-hour history severe abdominal pain that radiates to his back. His temperature is 39°C (103°F). Physical examination shows rebound tenderness. Laboratory studies show increased levels of serum amylase and lipase. He is diagnosed with inflammation of the organ at the tip of the arrow in the attached image. Which of the following organs most likely receives direct blood supply from the artery running on the superior aspect of the affected organ?

- A. Duodenum
- B. Liver
- C. Stomach
- D. Ascending colon
- E. Ileum



21. A 45-year-old man comes to the physician because of a 2-week history of abdominal pain that worsens after meals. Physical examination shows signs of malnutrition. Imaging studies show ischemia of the parts of the GI that receives blood supply from the artery at the pointer. Which of the following is the most likely embryological origin of the affected parts of the GI?

- A. Foregut
- B. Proctodeum
- C. Midgut
- D. Hindgut
- E. Dorsal mesentery
- F. Ventral mesentery



22. A 60-year-old man is brought to the emergency department by his wife because of a 12-hour history of passing bloody stools. He has a 10-year history of alcoholism and has been previously treated for hemorrhoids and jaundice. Physical examination shows dilated veins radiating from his umbilicus. Rectal exam shows the stool contains bright red blood. The anastomosis of which of the following veins is most likely responsible for the patient's presenting complaint?

- A. Left gastric and esophageal
- B. Middle and inferior rectal
- C. Superior and middle rectal
- D. Inferior rectal and pudendal
- E. Paraumbilical and superficial epigastric

23. A 60-year-old man is admitted to the hospital for treatment of a posterior abdominal wall tumor that has resulted in lesions that have invaded the superior mesenteric periarterial plexus. Which of the following nervous pathways is most likely affected?

- A. Preganglionic parasympathetics to the posterior aspect of the stomach
- B. Postganglionic sympathetics to the ascending colon
- C. Visceral afferents from the duodenal bulb
- D. Postganglionic sympathetics from the descending colon
- E. Visceral afferents from the sigmoid colon

24. A 55-year-old woman is admitted to the hospital for treatment of a pancreatic tumor. A diagnostic arteriogram shows that the neoplasm is in the tail of the pancreas and has compromised the blood supply to another organ. Which of the following organs is most likely deprived of blood?

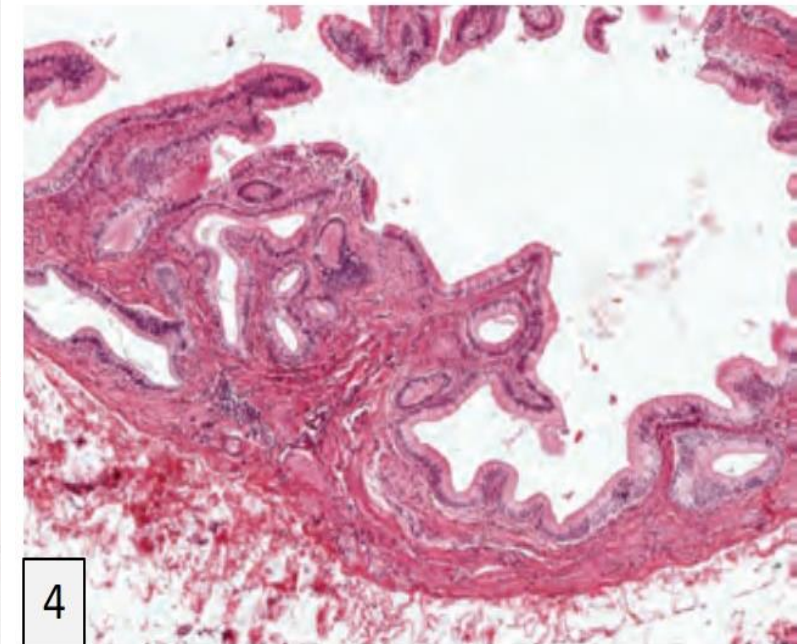
- A. Duodenum
- B. Gallbladder
- C. Kidney
- D. Liver
- E. Spleen

25. A 38-year-old woman comes to the emergency department because of a 3-day history of nausea, vomiting and intermittent “colicky” abdominal pain. She says her stool is “clay-colored” and her urine is dark. Physical examination shows an ill looking patient with yellowing of the skin. There is also right upper quadrant tenderness on deep palpation. Laboratory tests show elevated total bilirubin, AST, ALT and ALP enzymes. Lipase and amylase enzymes are within normal range. Ultrasound shows large gallstones in the biliary ductal system. Which of the following cells lines the structure that is most likely blocked by the gallstones that explain this patient’s clinical presentation?

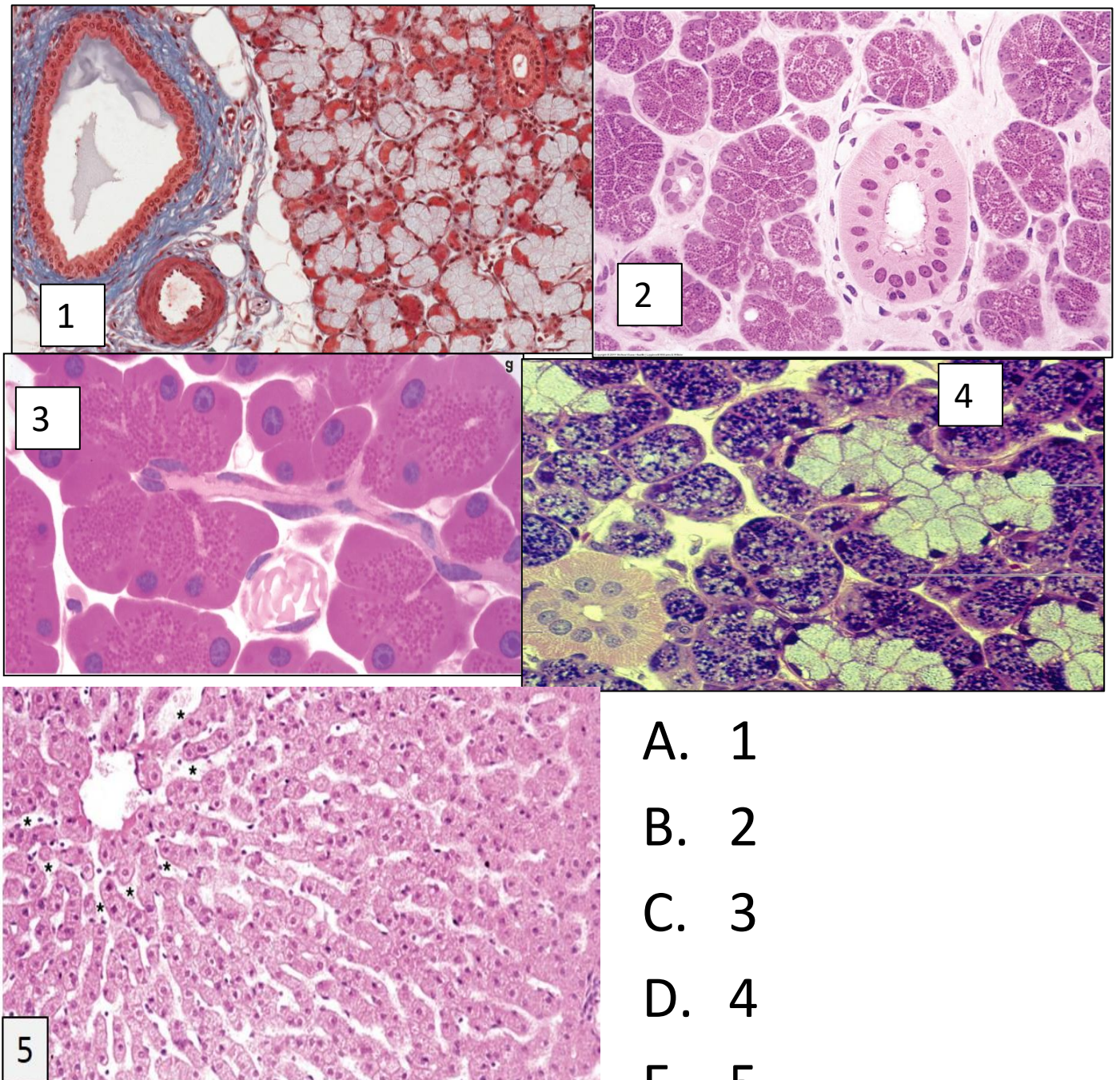
- A. Hepatocytes
- B. Cholangiocytes
- C. Serous acinar cells
- D. Kupfer cells
- E. Columnar absorptive cells
- F. Ito cells
- G. Sinusoidal endothelial cells

26. A 53-year-old man comes to the emergency department because of a 4-hour history of severe abdominal pain and nausea. The pain started around the umbilical region and now intensifies and shifted to the right iliac fossa. Physical examination shows positive psoas test and rebound tenderness at McBurney's point. Which of the following is slides best describes the inflamed organ?

- A. 1
- B. 2
- C. 3
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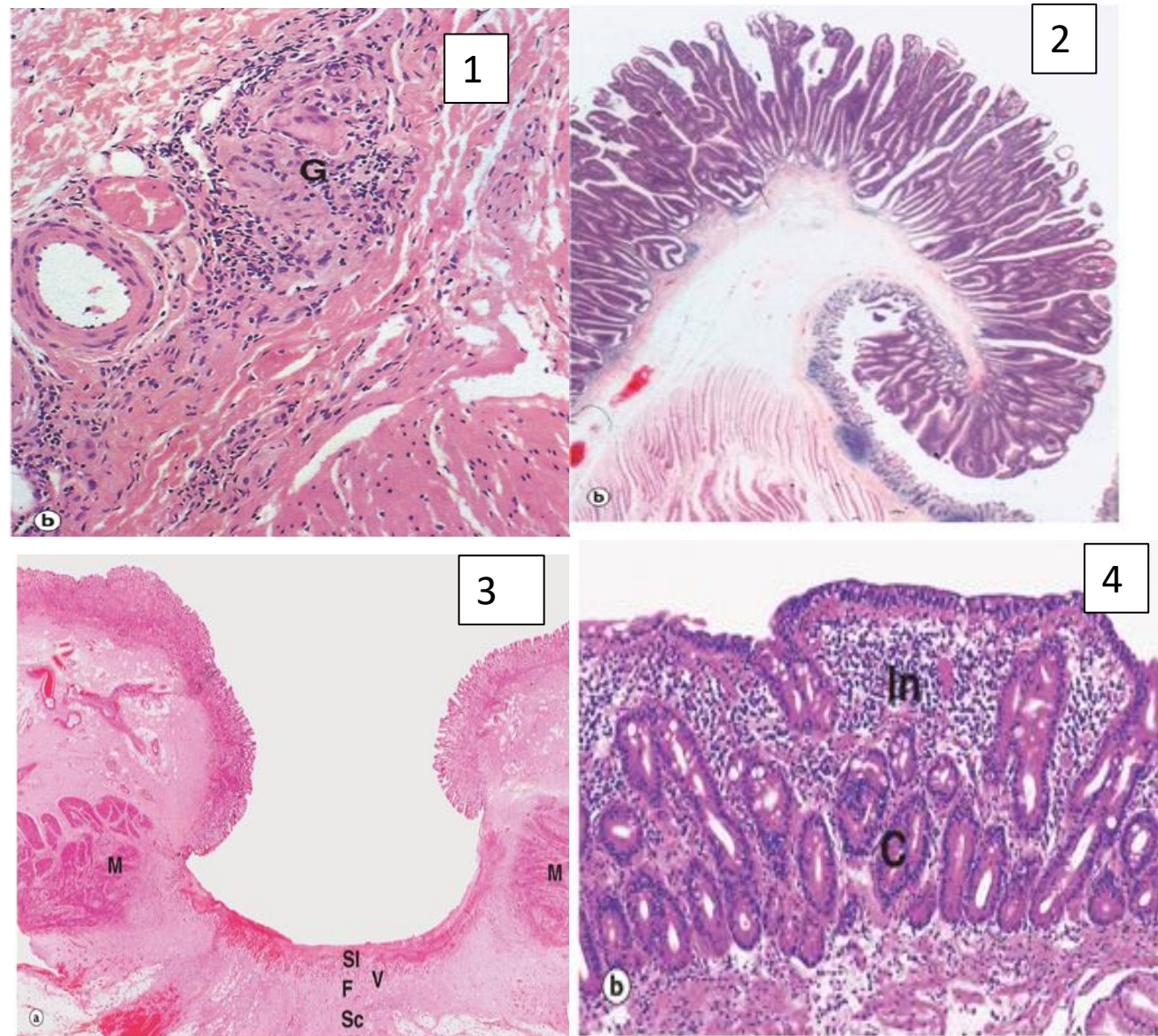
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- A. 1
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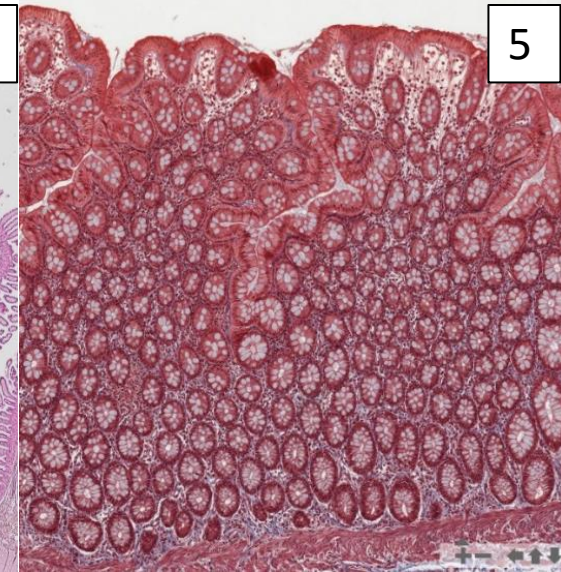
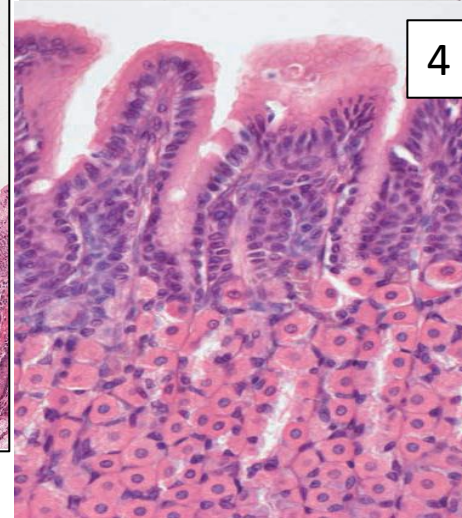
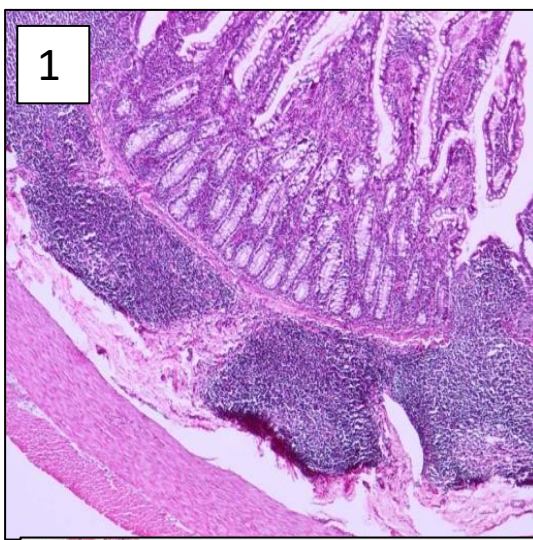
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- A. 1
- B. 2
- C. 3
- D. 4



29. A 28-year-old woman comes to the physician because of a 1-year history of recurrent diarrhea, abdominal pain, fever and malnutrition. Endoscopy shows inflammation of the part of the bowel located 2 feet (~60cm) distal to the stomach. Which of the following is most likely the area affected?

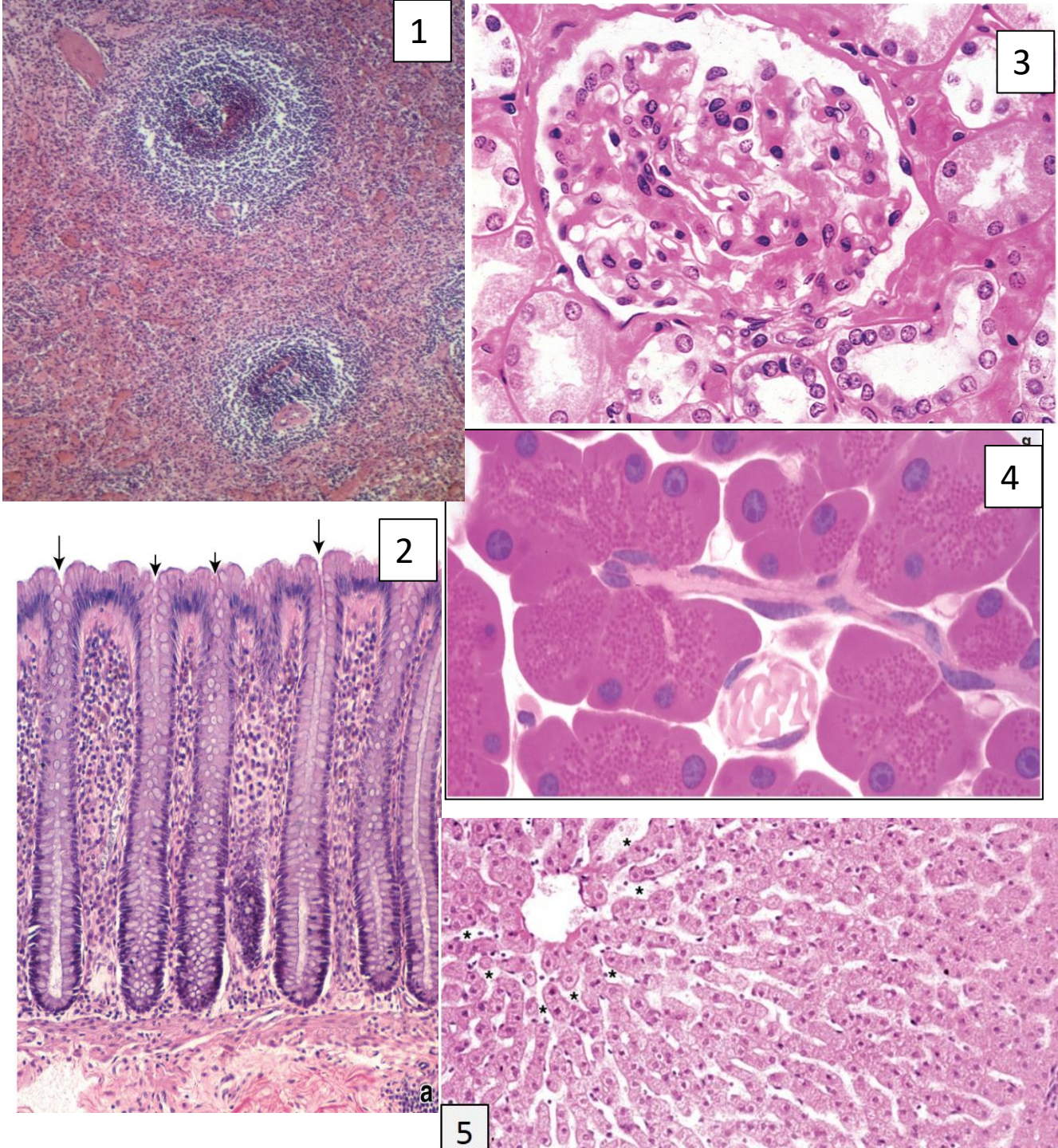
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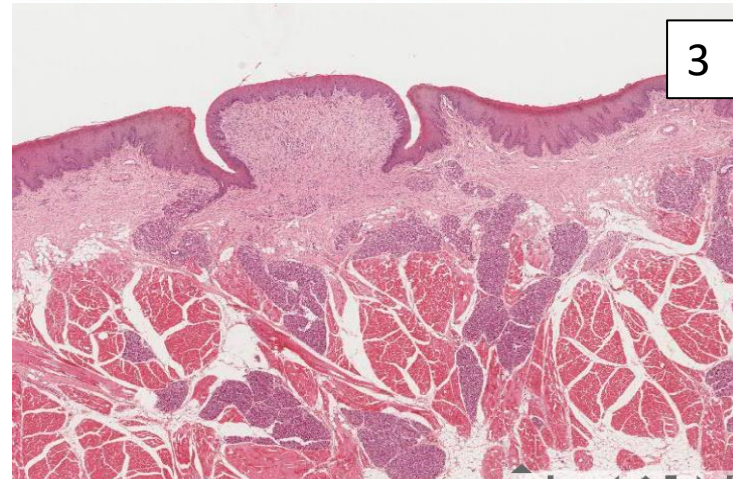
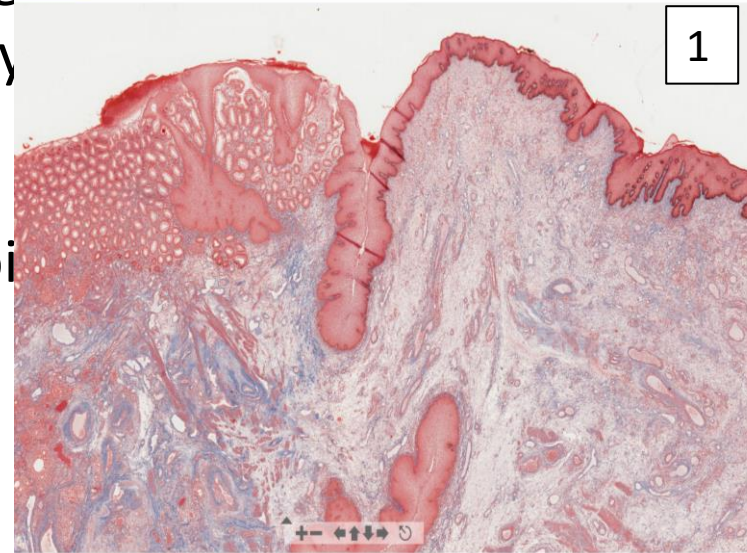
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- E. 5

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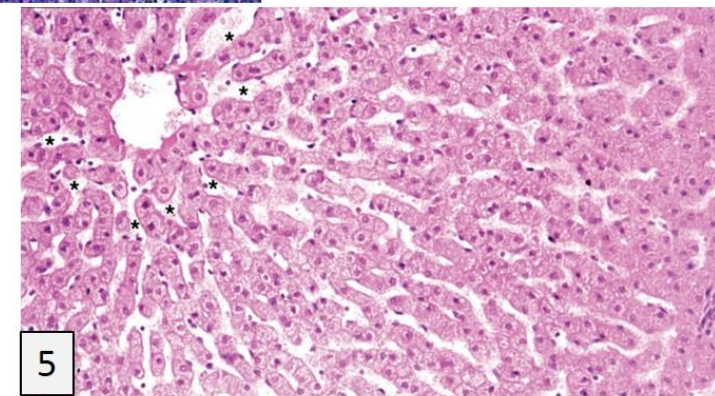
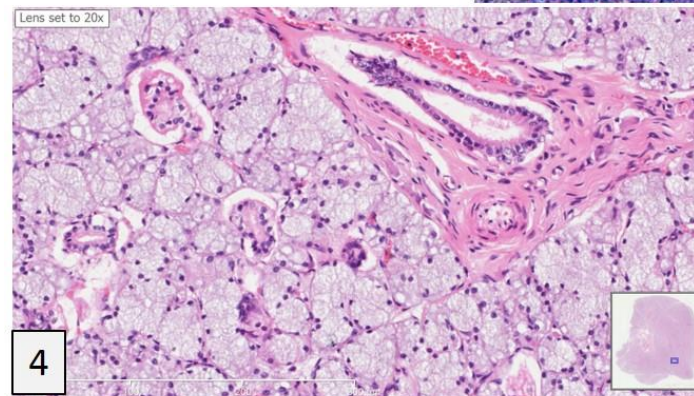
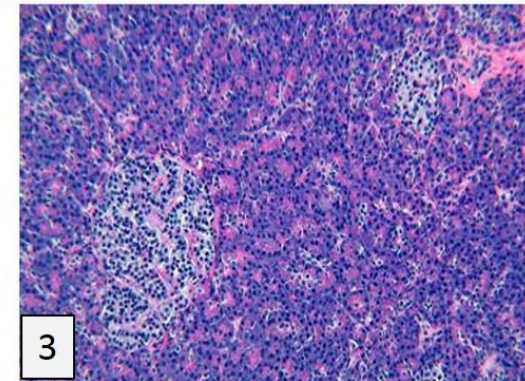
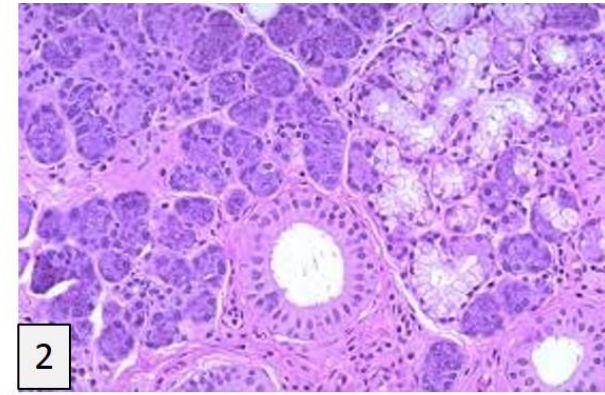
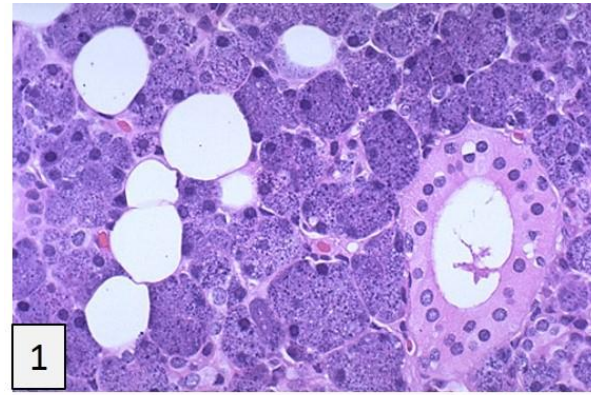
31. A 57-year-old woman comes to the physician because of a 6-month history of irregular bowel habits and blood in the stool. Physical examination shows abdominal tenderness and macroscopic rectal bleeding. Laboratory studies of the blood show no abnormalities. Colonoscopy shows a 1.5 cm mass in the distal part of the GI tract at anal canal. Which of the following images best corresponds to this region?



- A. 1
- B. 2
- C. 3
- D. 4

32. A 64-year-old man comes to the physician because of a 3-month history of a swelling in the floor of the mouth. Physical examination shows a firm, 3 cm mass in sublingual salivary gland. Specimens are shown in the attached images. Which of the following images best corresponds to the affected gland?

- A. 1
- B. 2
- C. 3
- D. 4
- E. 5



Key

1. F
2. B
3. C
4. D
5. C
6. A
7. C
8. A
9. C
10. F
11. C
12. D
13. D
14. A
15. B
16. C

17. C
18. D
19. D
20. C
21. D
22. C
23. B
24. E
25. B
26. C
27. C
28. D
29. C
30. B
31. A
32. D